

LambdaEta yield from production-vertex missing mass

Macro: plot\_lambda\_eta\_yield.cc | Executed: 2026-10-01 12:24:21

Beam momentum setting: 735 MeV/c

Runs used: 02447, 02449, 02450, 02451, 02452, 02453, 02454, 02456, 02457, 02458, 02459, 02460, 02462,

Input: /gpfs/home/had/haein/data/JPARC2025Nov\_root/physics-735/runXXXXX\_DstTPCLambdaEta.root

Selection: X\_prod\_found; Lambda mass window  $1.09 < M(p \pi^-) < 1.13 \text{ GeV}/c^2$

Lambda mass displayed: 1.04 to 1.20  $\text{GeV}/c^2$ ; peak fit: 1.10 to 1.13  $\text{GeV}/c^2$

Pre-cut Lambda peak fit model: Gaussian + linear background, 1.100 to 1.130  $\text{GeV}/c^2$

Production vertex LH2 cut: off

Pion selection: lambda\_pion\_track\_id matched to pion\_beam\_window\_status == 0 when branch exists

Combined eta fit: Gaussian + linear background up to kinematic endpoint, 0.50 to 0.60  $\text{GeV}/c^2$

Linear background is included only below the  $K^+p \rightarrow \Lambda X$  kinematic endpoint

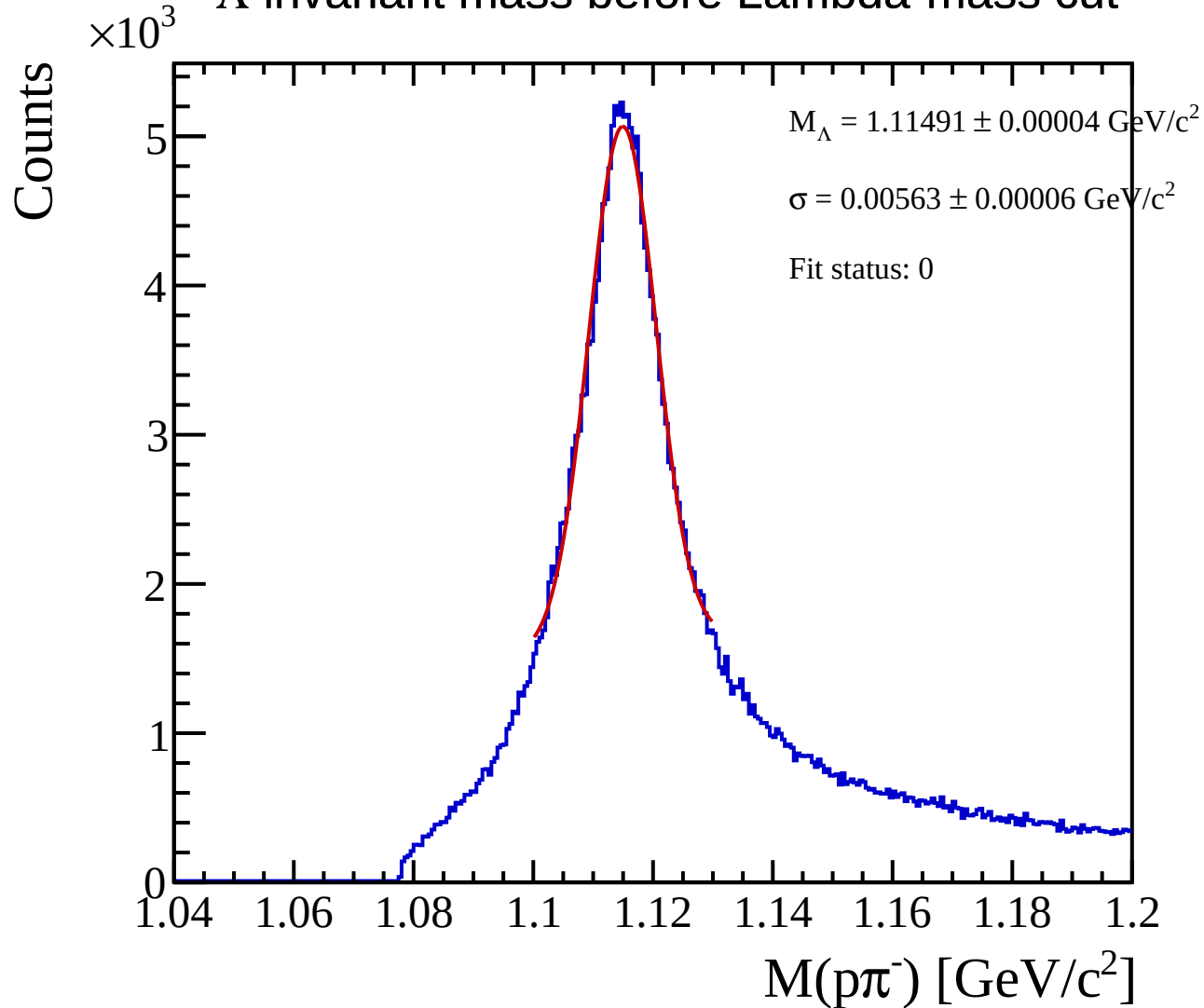
Selected-candidate mean beam momentum = 0.7518  $\text{GeV}/c$ ; kinematic endpoint = 0.5612  $\text{GeV}/c^2$

Momentum-dependent eta yield: Gaussian signal area from the combined fit

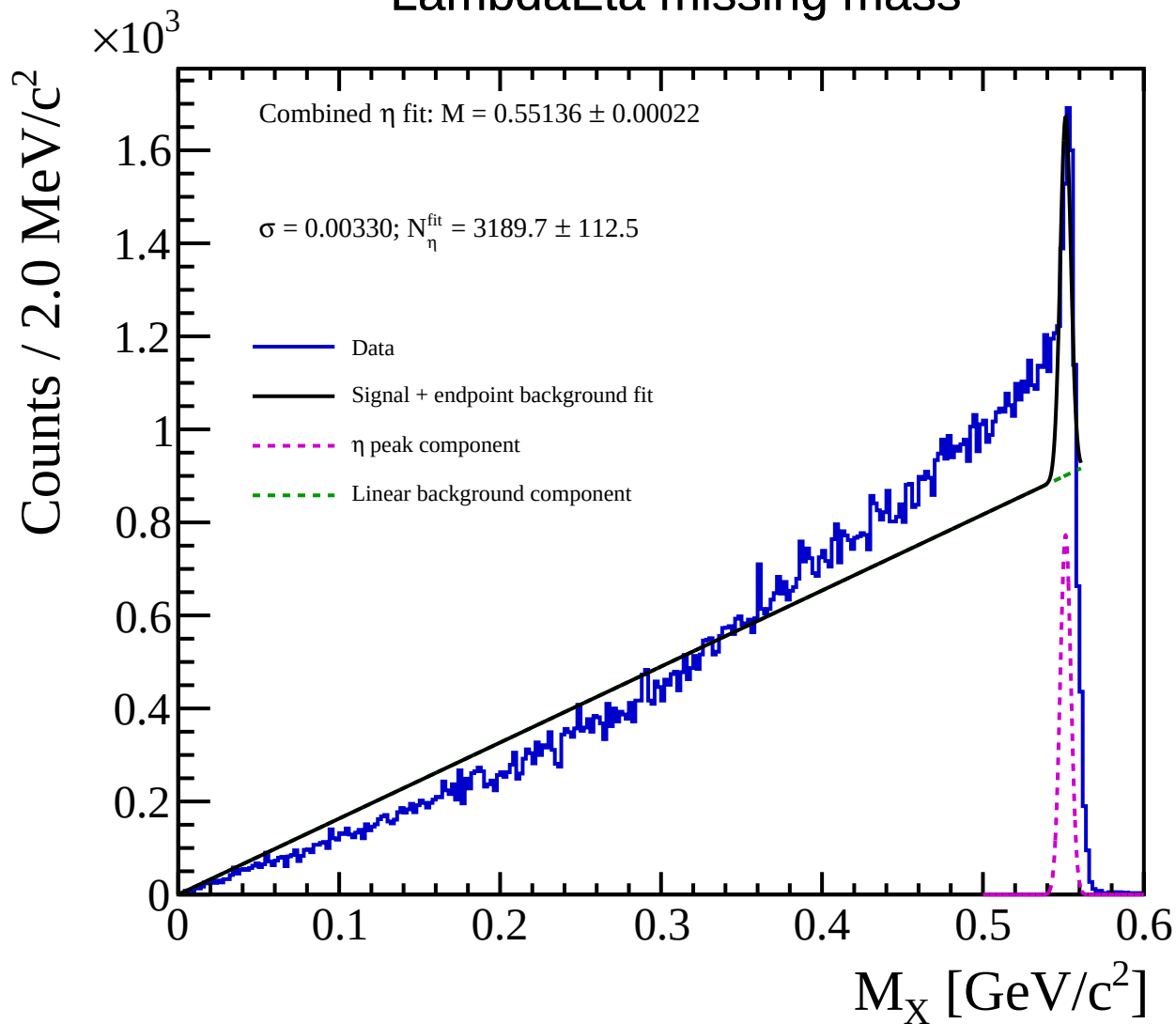
Production beam momentum: 25 bins x 4  $\text{MeV}/c$  from 700 to 800  $\text{MeV}/c$

Events read: 5659492 | candidates checked: 383144 | pre-cut Lambda entries: 367306 | eta selection: 212572

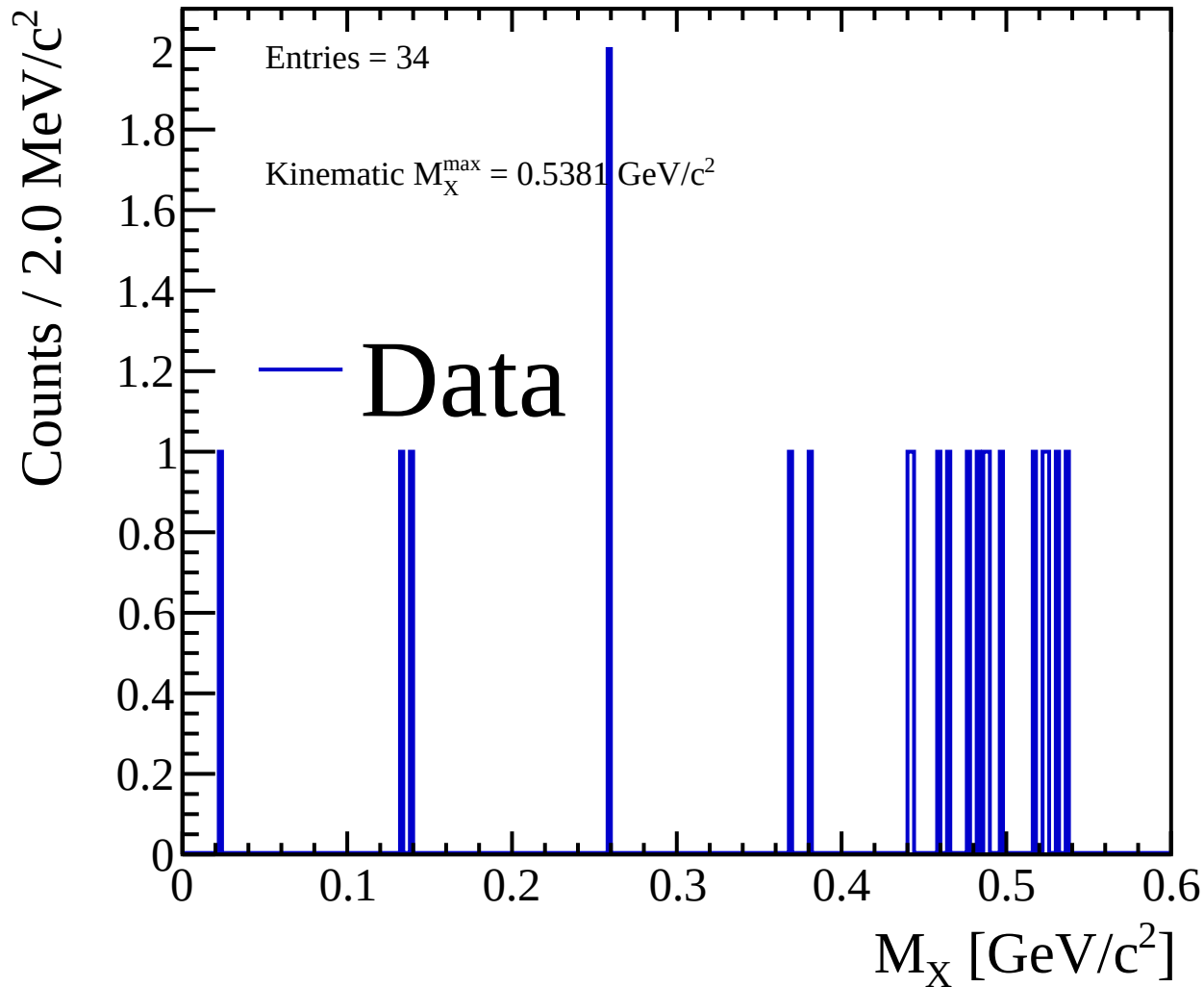
# $\Lambda$ invariant mass before Lambda-mass cut



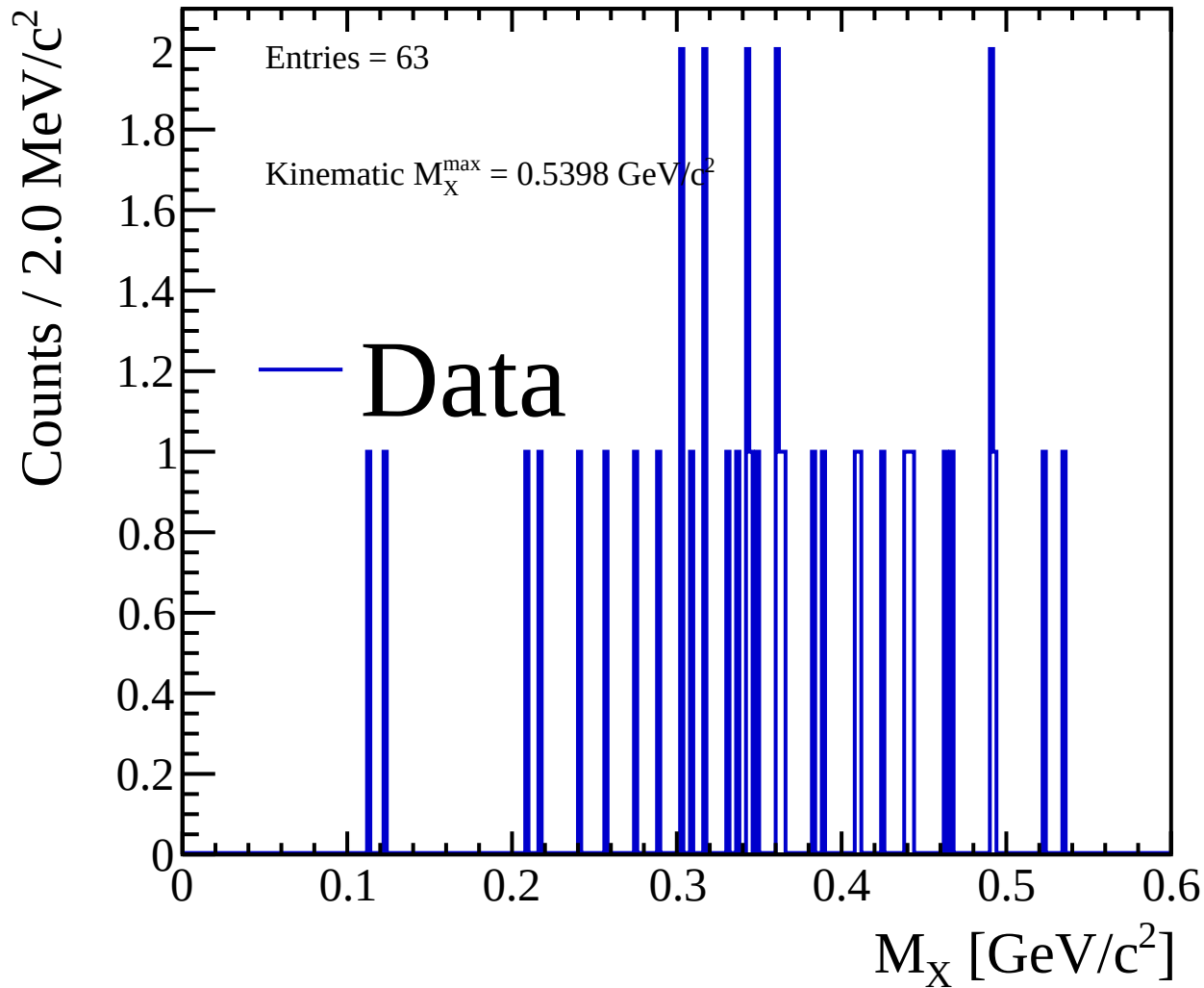
# LambdaEta missing mass



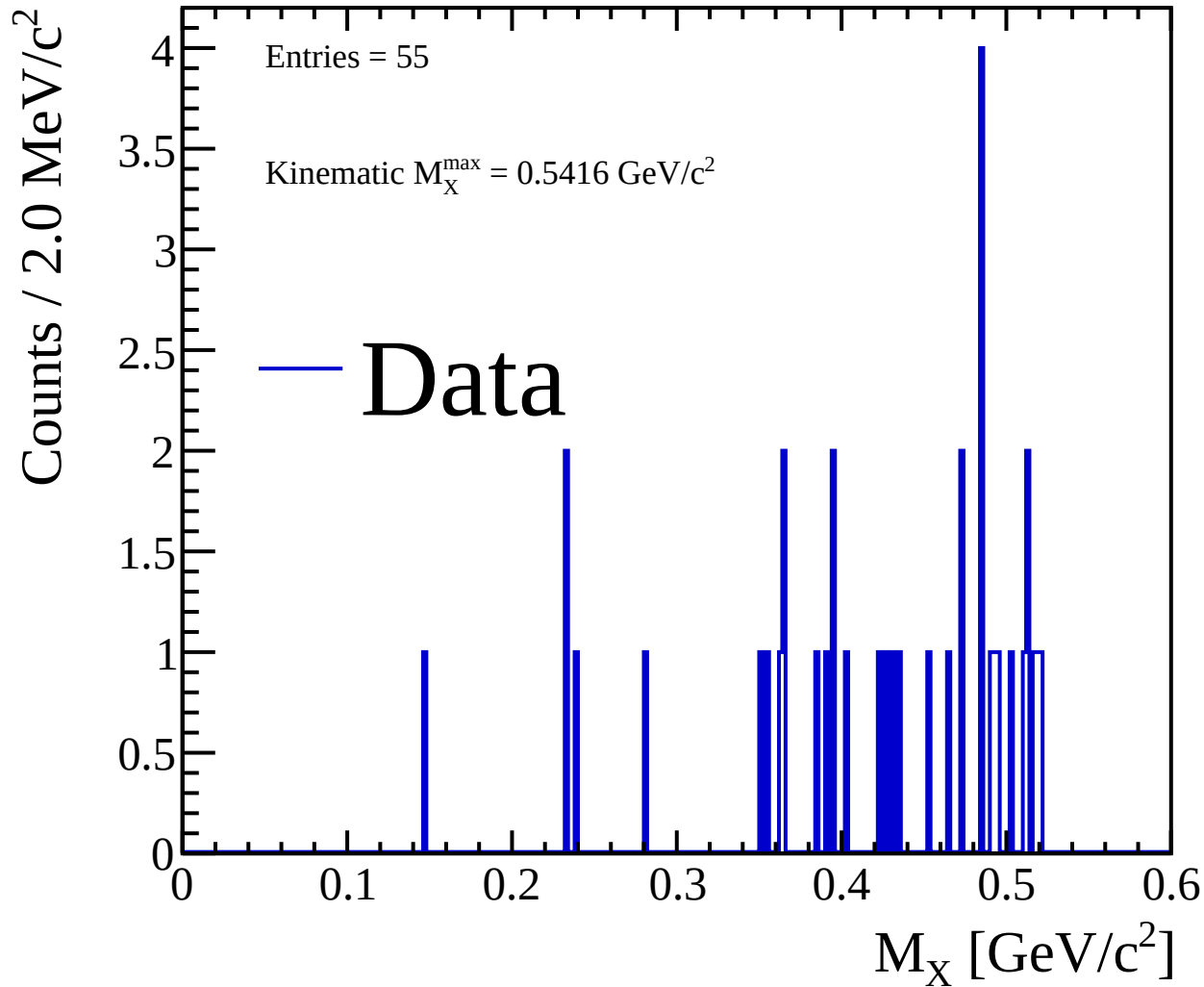
Production-vertex  $K^-$  momentum 700--704 MeV/c



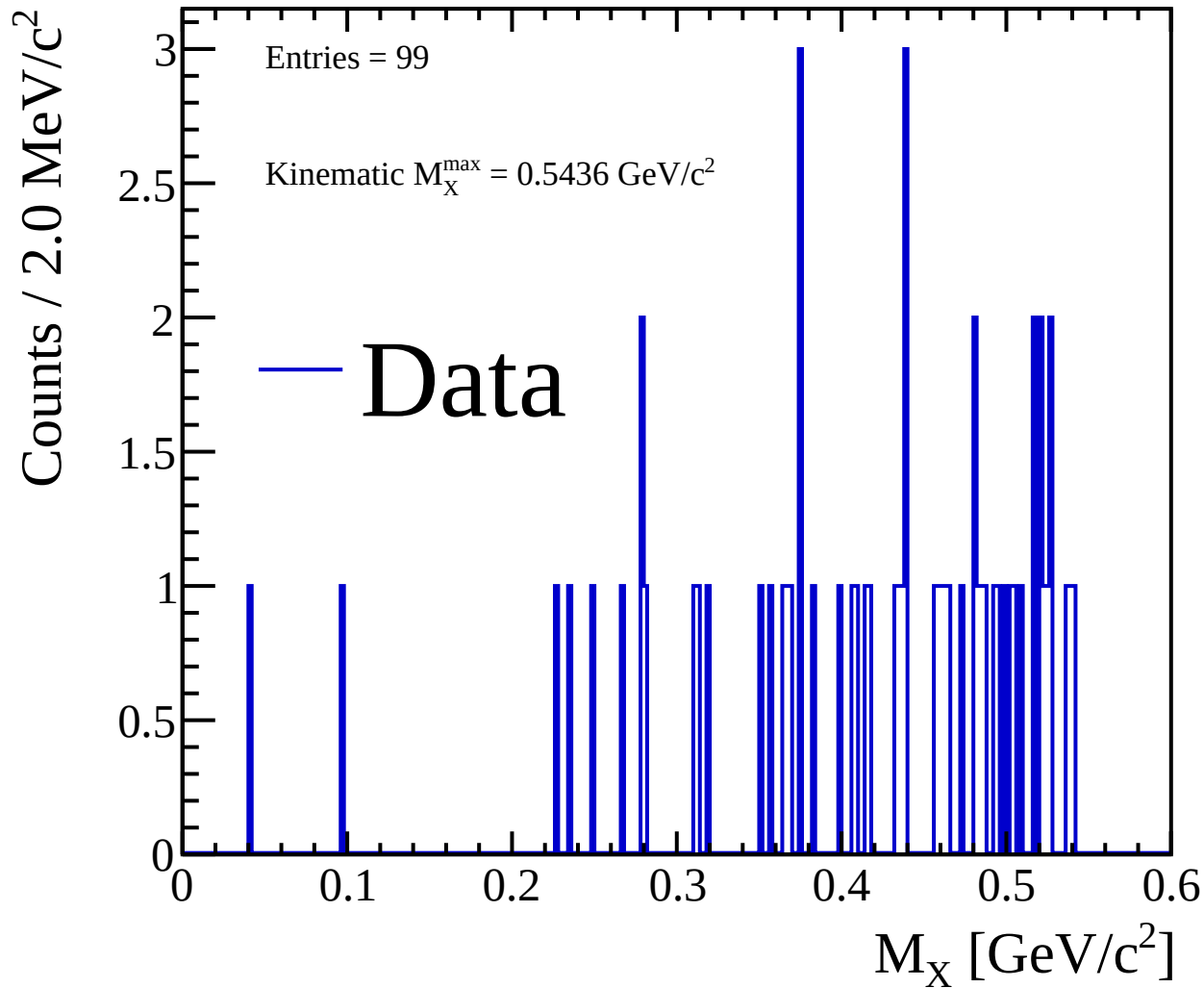
Production-vertex  $K^-$  momentum 704--708 MeV/c



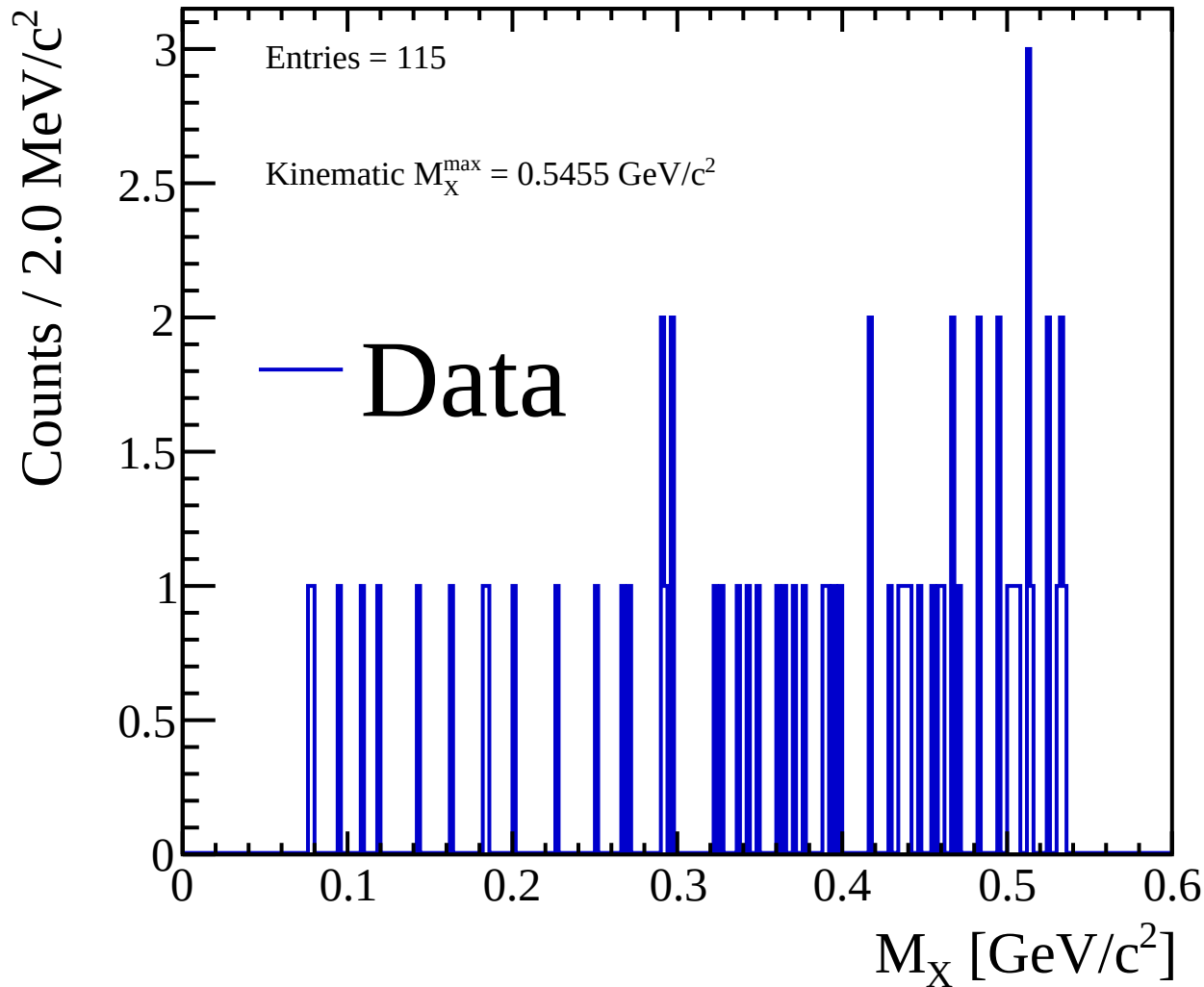
Production-vertex  $K^-$  momentum 708--712 MeV/c



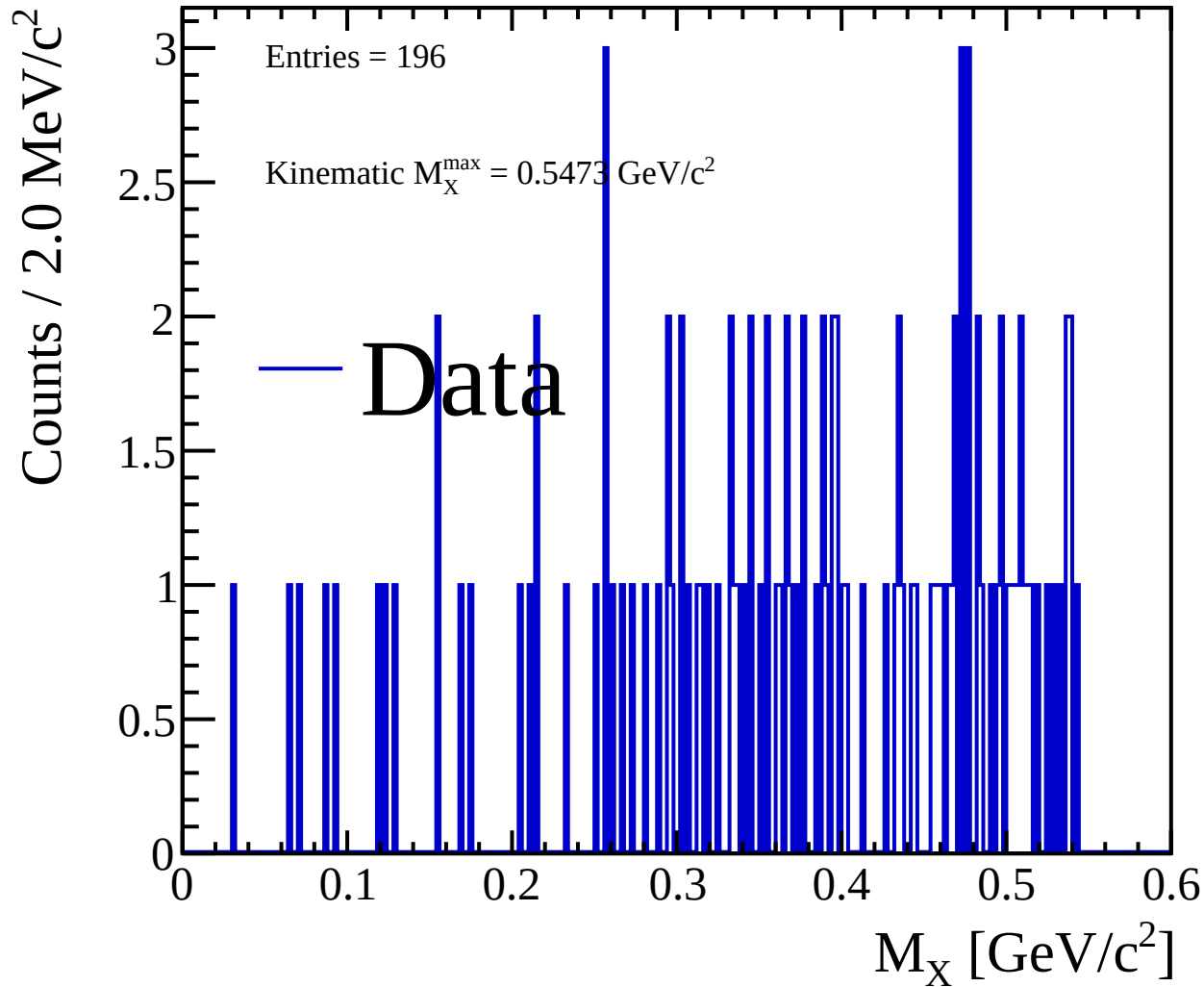
Production-vertex  $K^-$  momentum 712--716 MeV/c



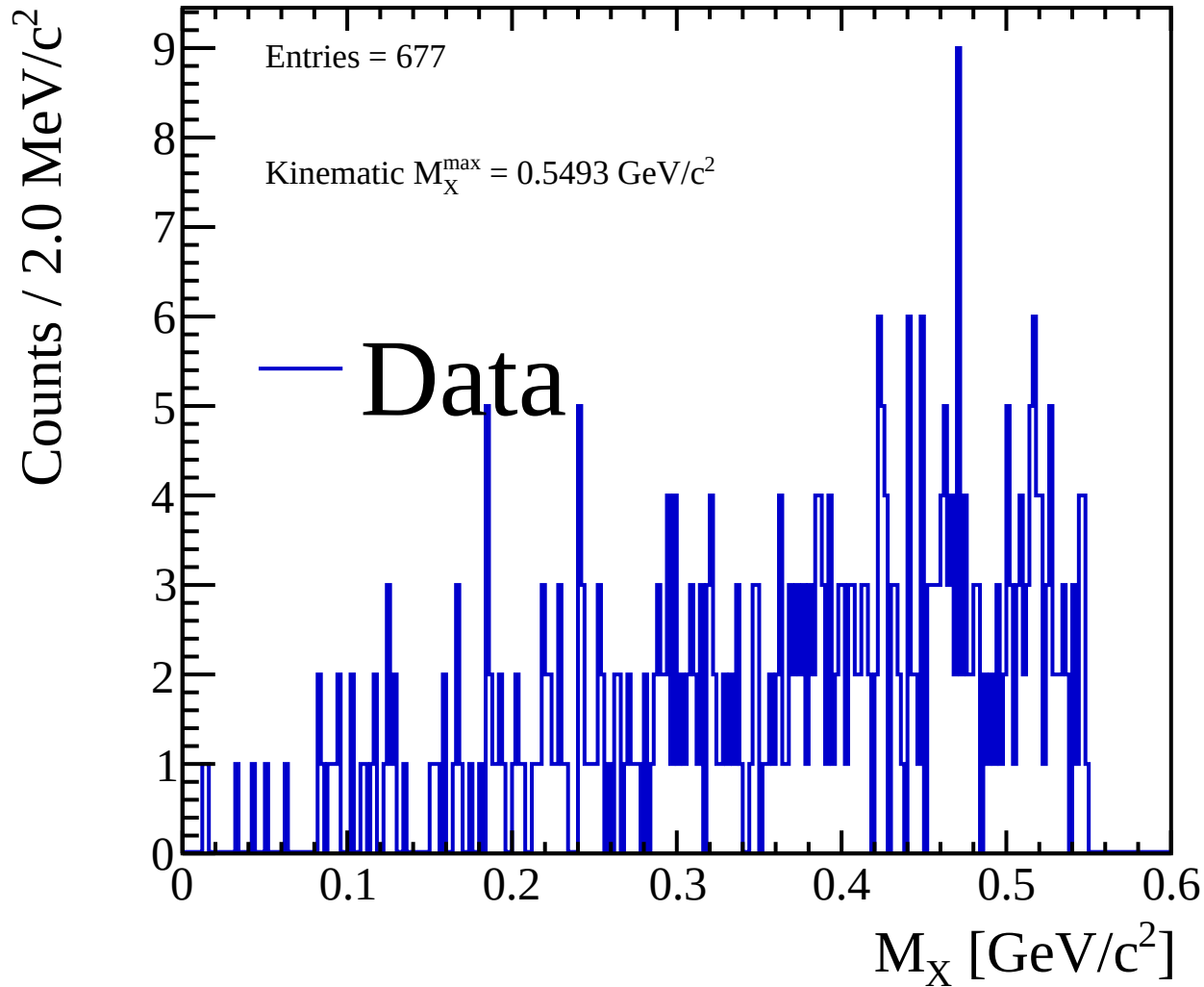
Production-vertex  $K^-$  momentum 716--720 MeV/c



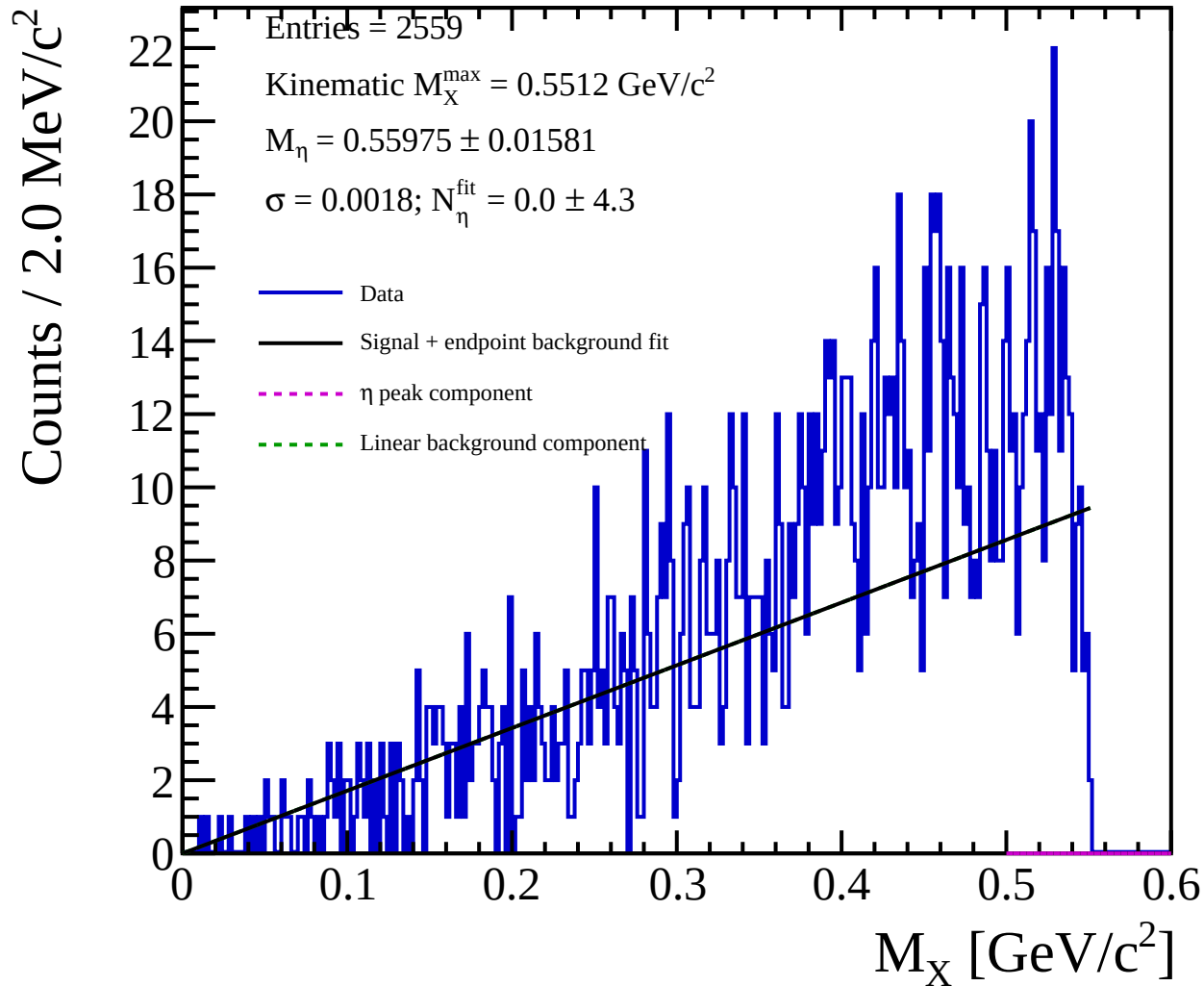
Production-vertex  $K^-$  momentum 720--724 MeV/c



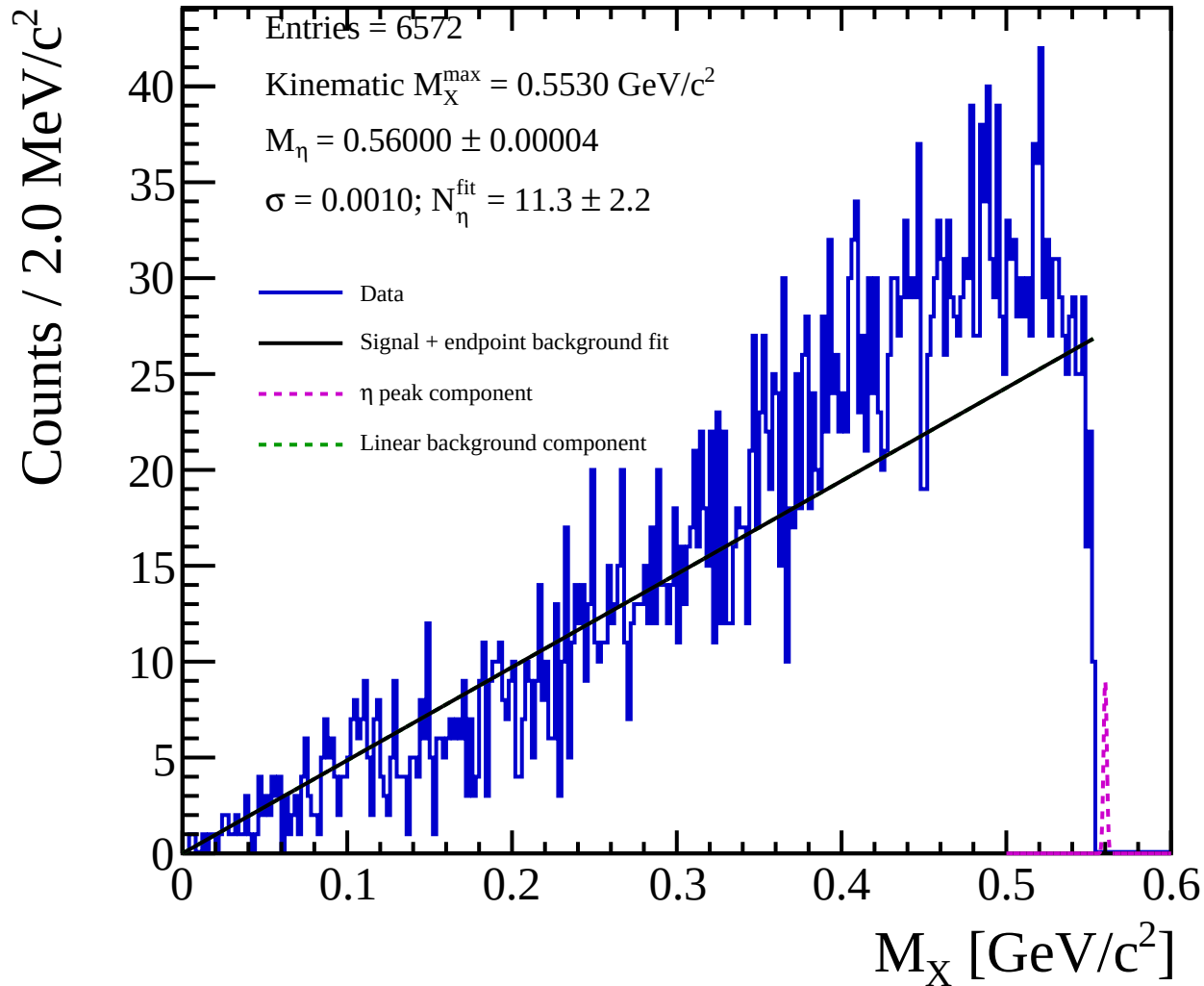
Production-vertex  $K^-$  momentum 724--728 MeV/c



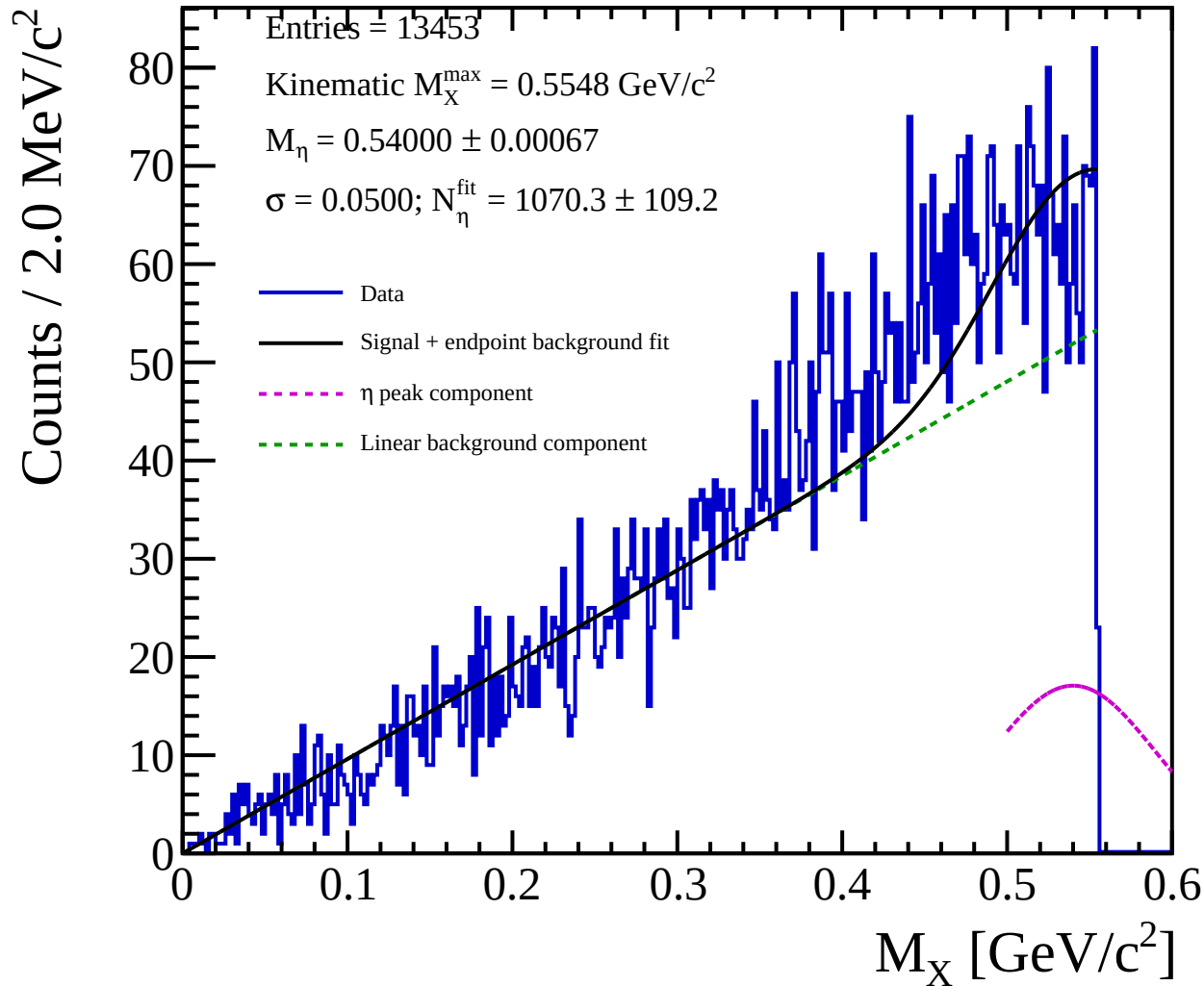
# Production-vertex $K^-$ momentum 728--732 MeV/c



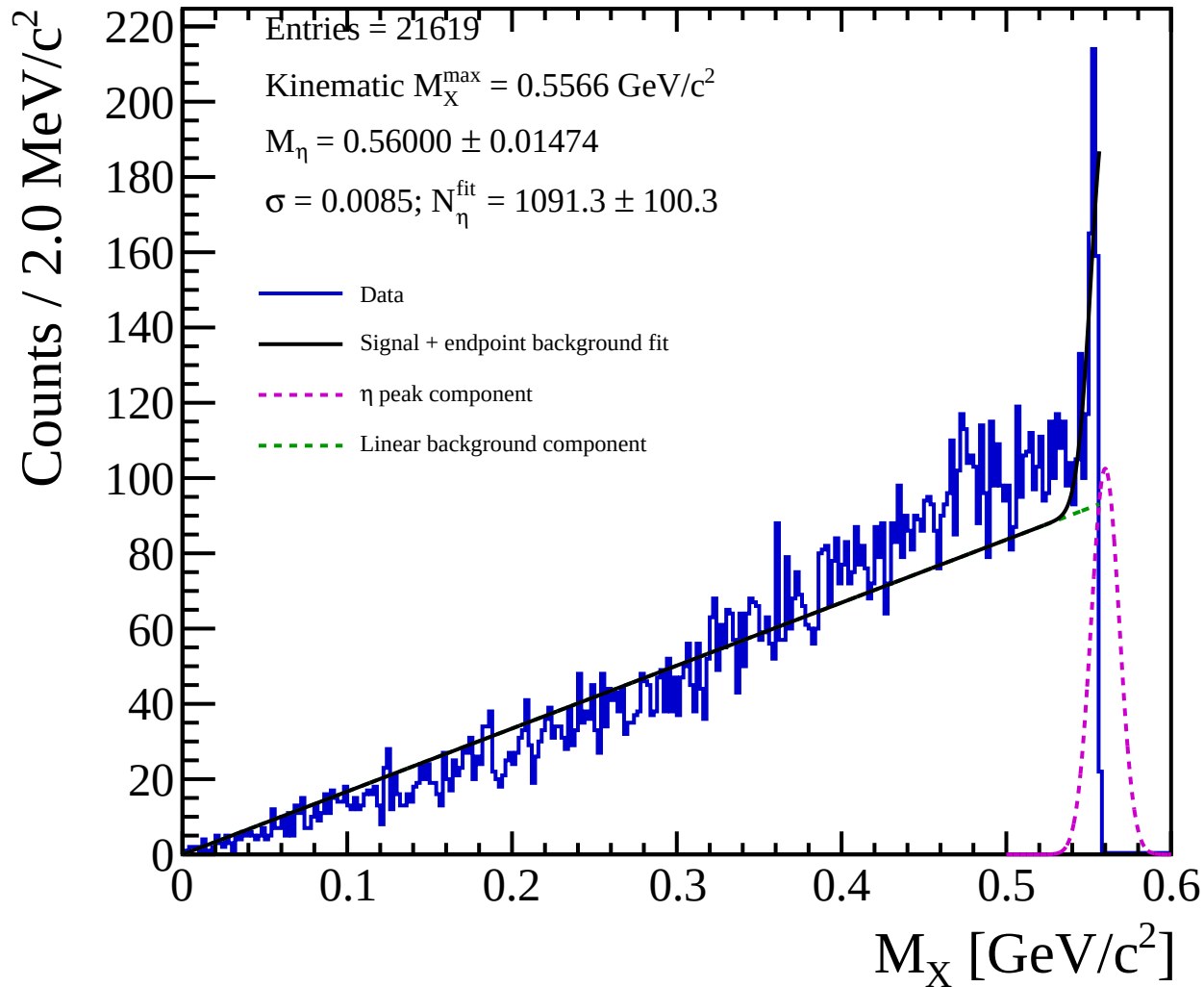
# Production-vertex $K^-$ momentum 732--736 MeV/c



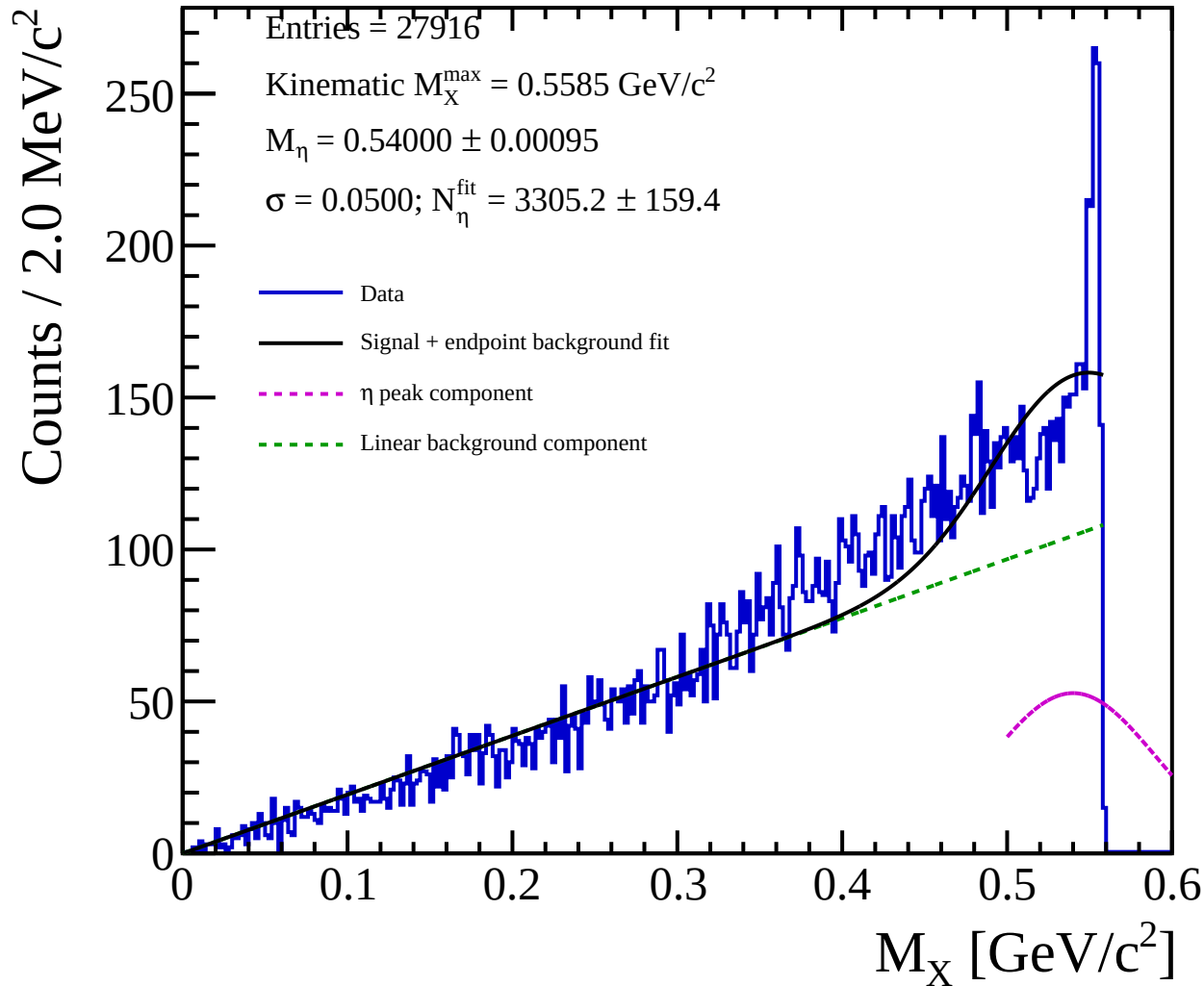
# Production-vertex $K^-$ momentum 736--740 MeV/c



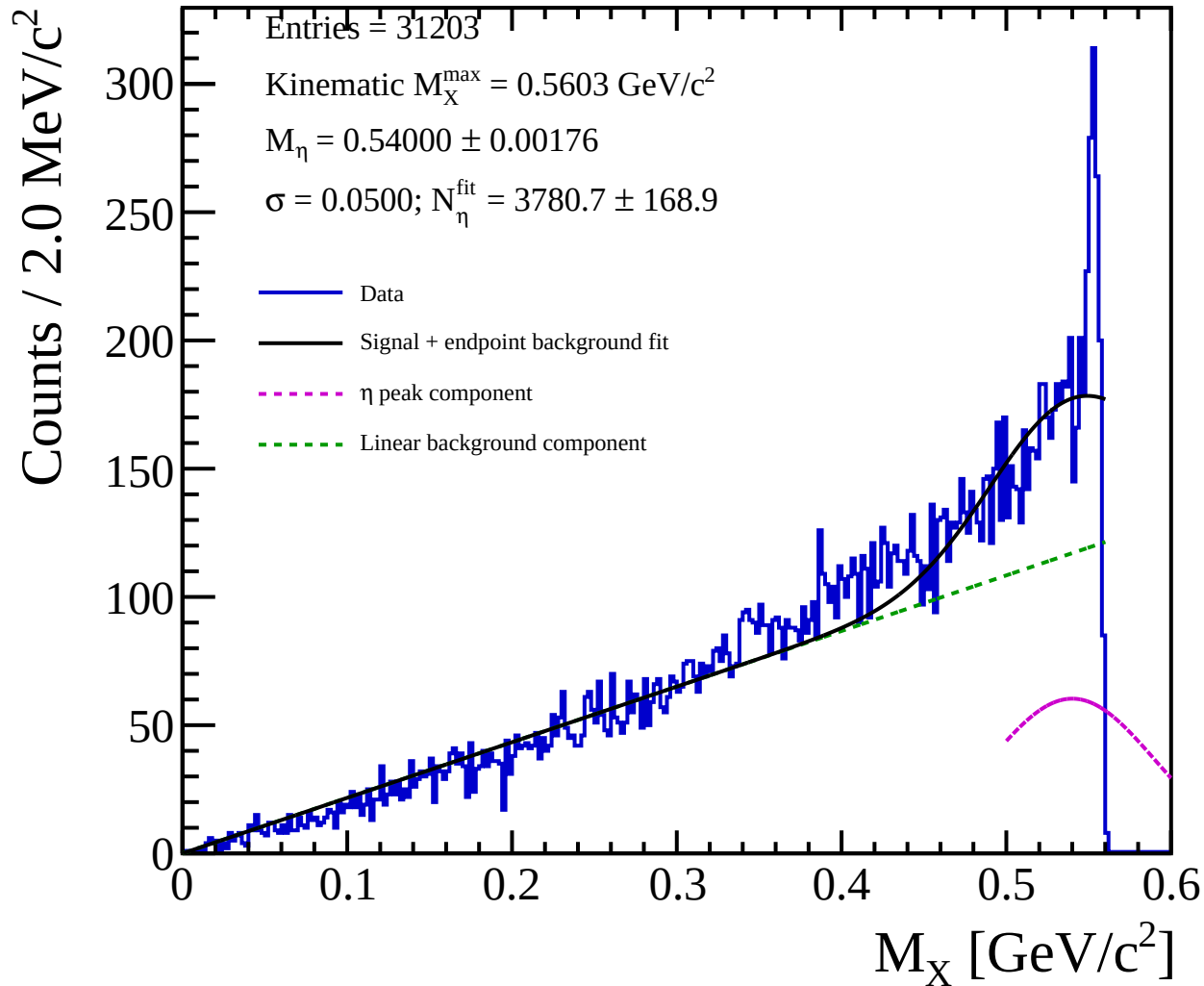
# Production-vertex $K^-$ momentum 740--744 MeV/c



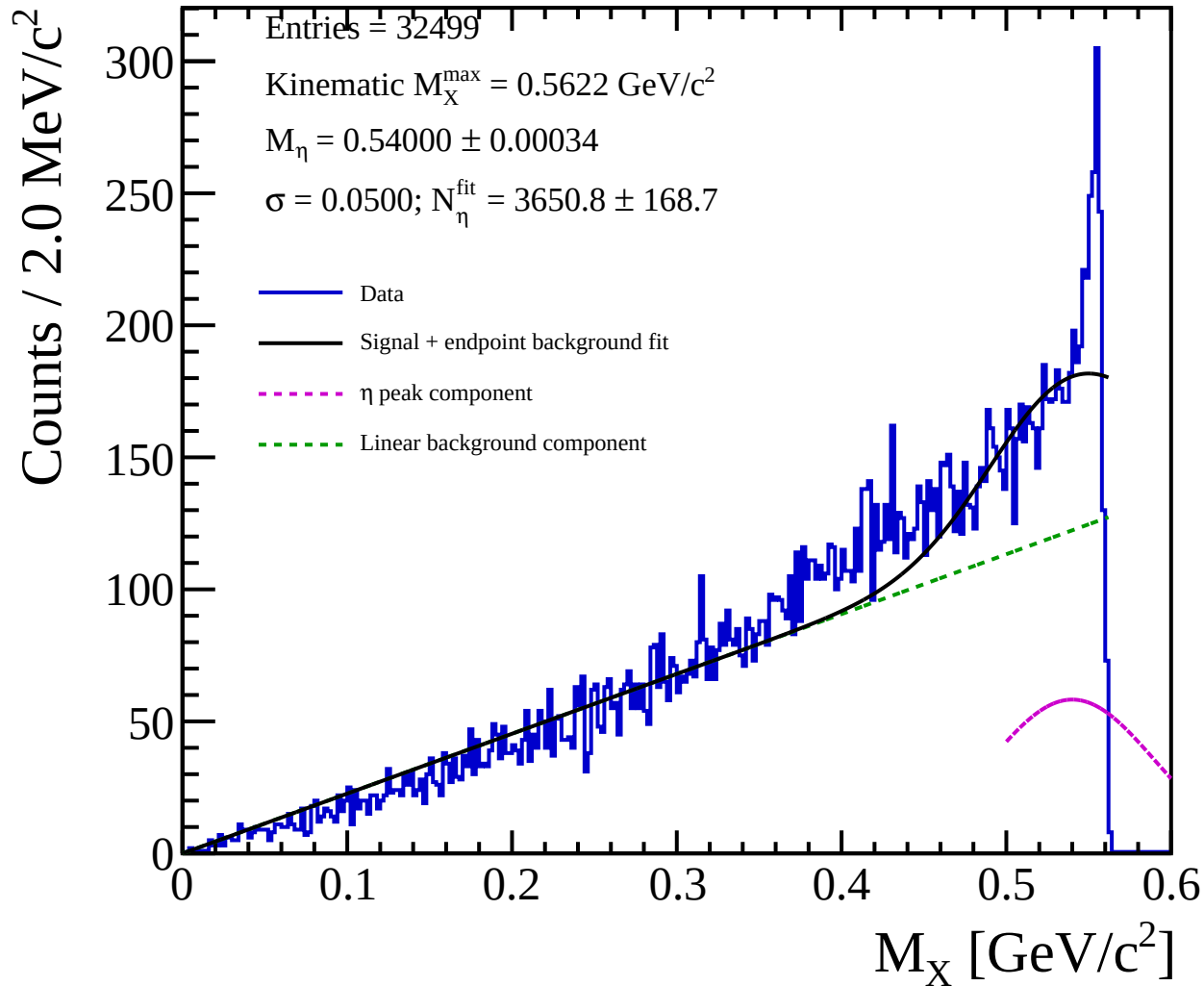
# Production-vertex $K^-$ momentum 744--748 MeV/c



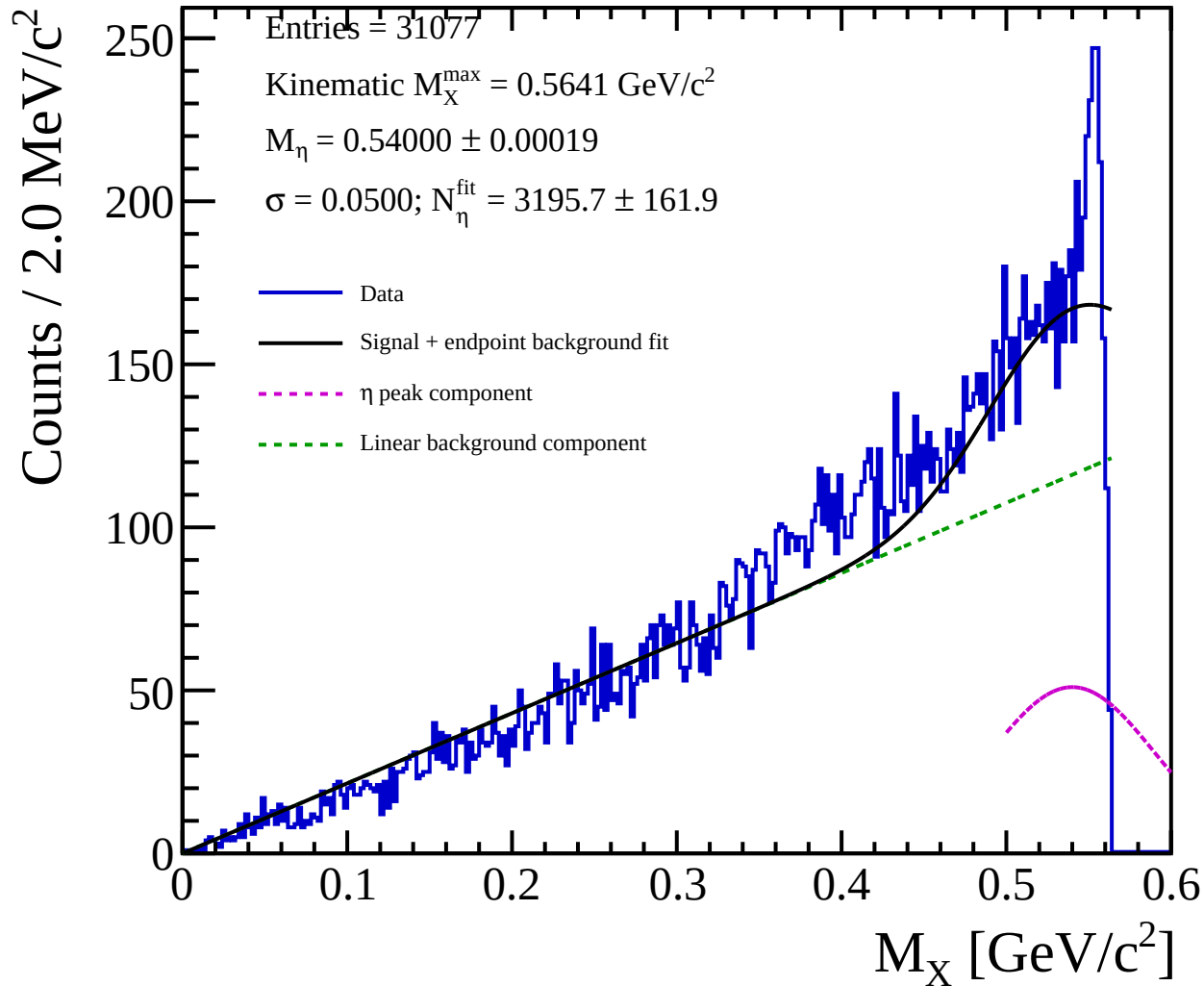
# Production-vertex $K^-$ momentum 748--752 MeV/c



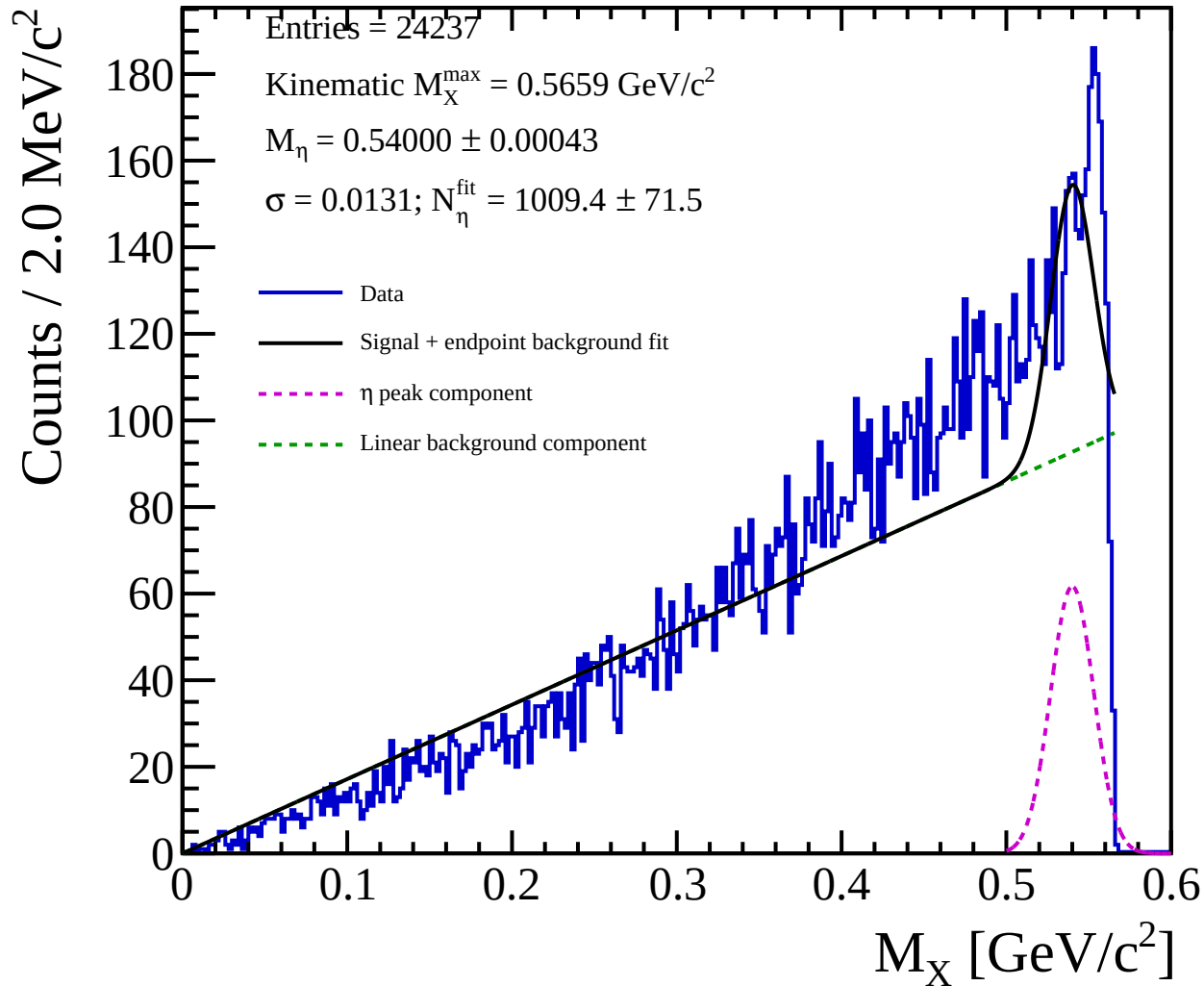
# Production-vertex $K^-$ momentum 752--756 MeV/c



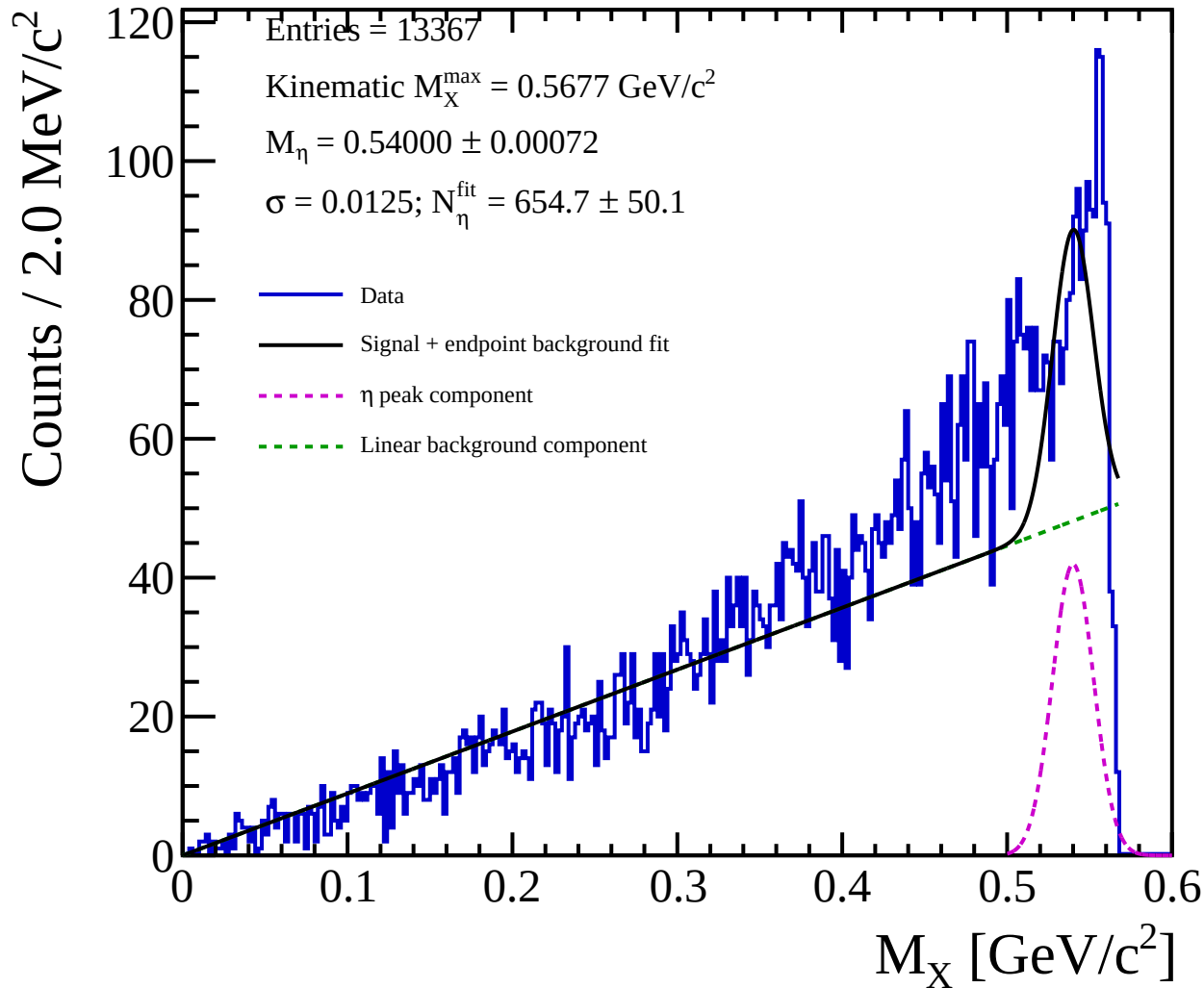
# Production-vertex $K^-$ momentum 756--760 MeV/c



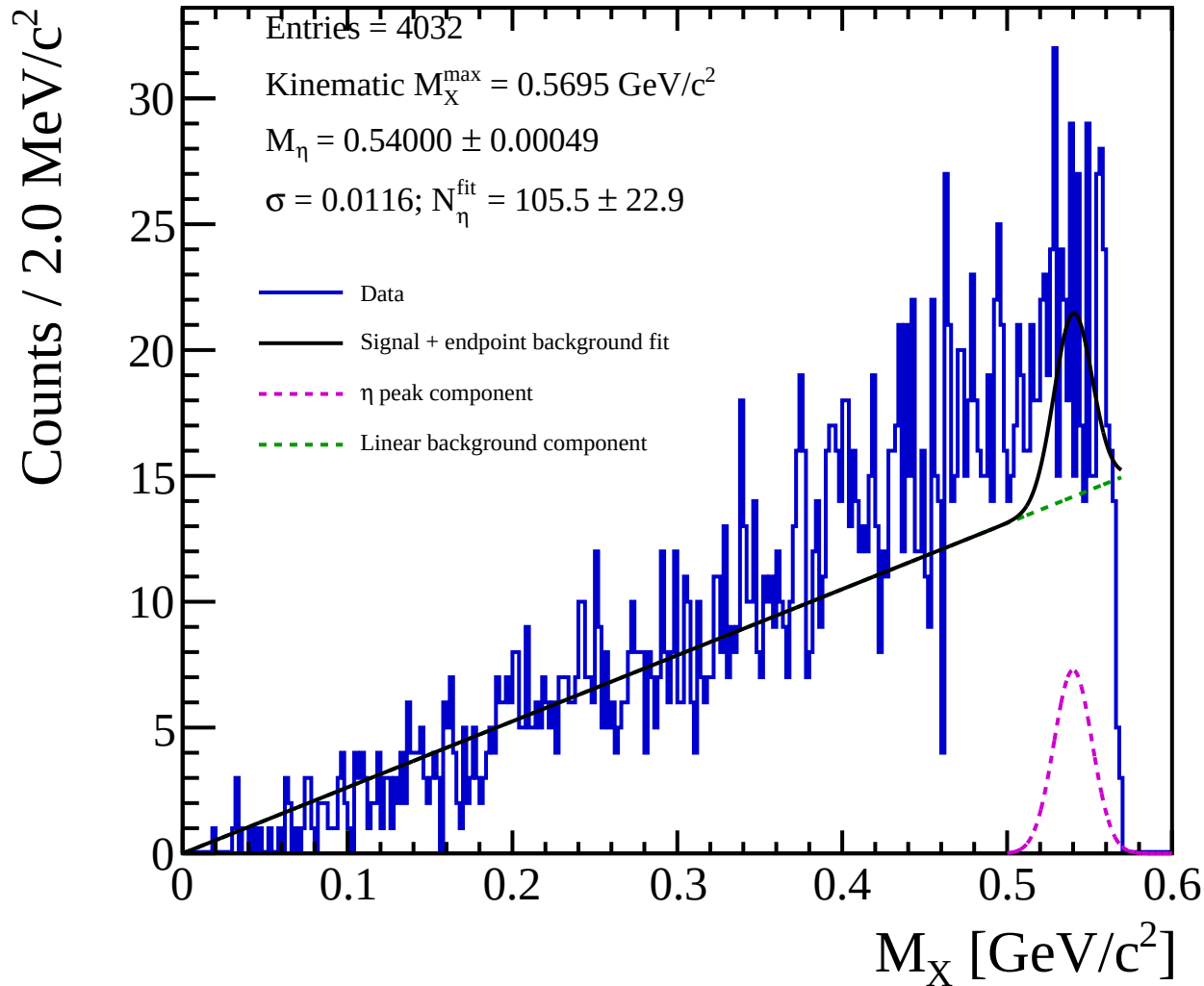
# Production-vertex $K^-$ momentum 760--764 MeV/c



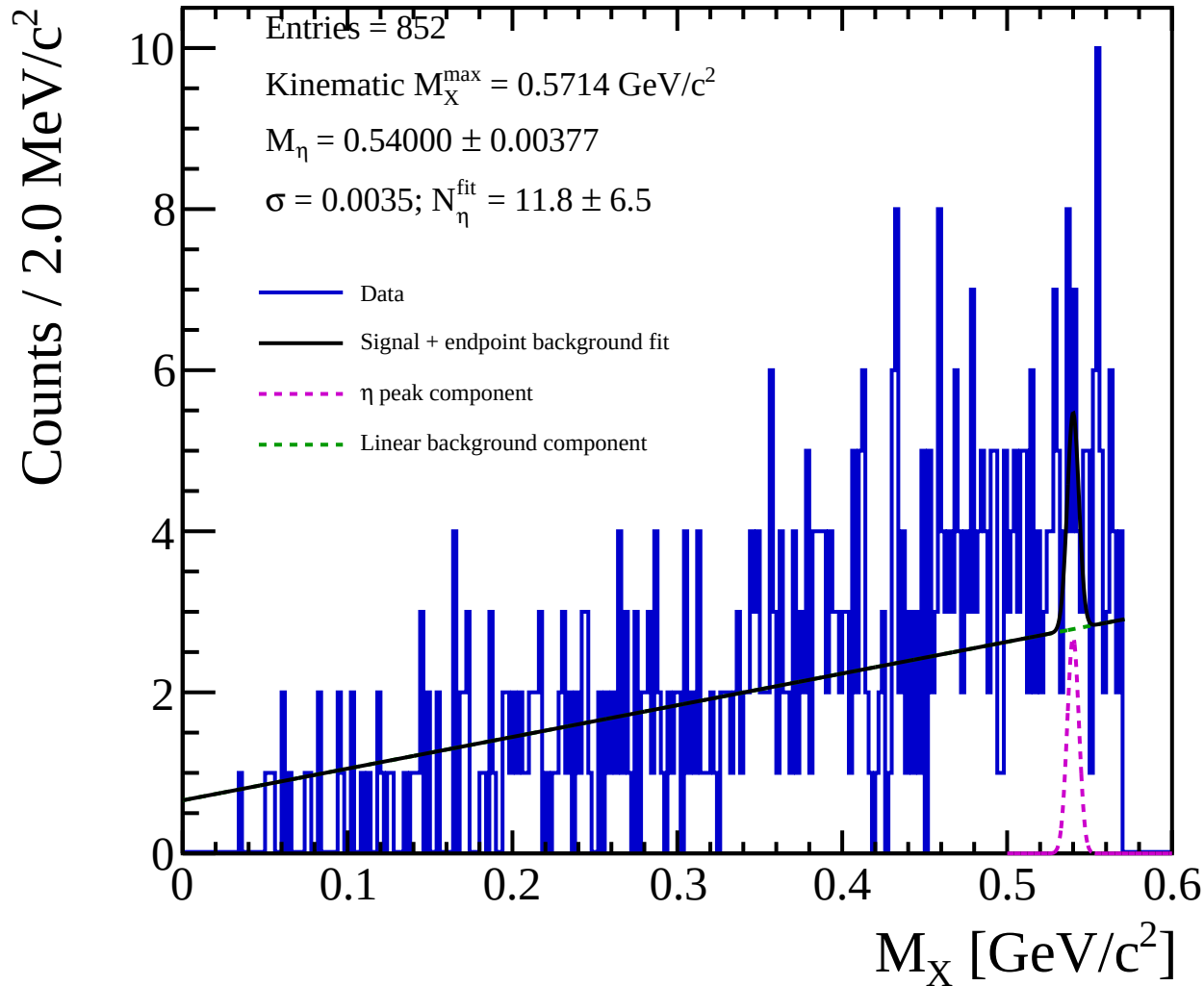
# Production-vertex $K^-$ momentum 764--768 MeV/c



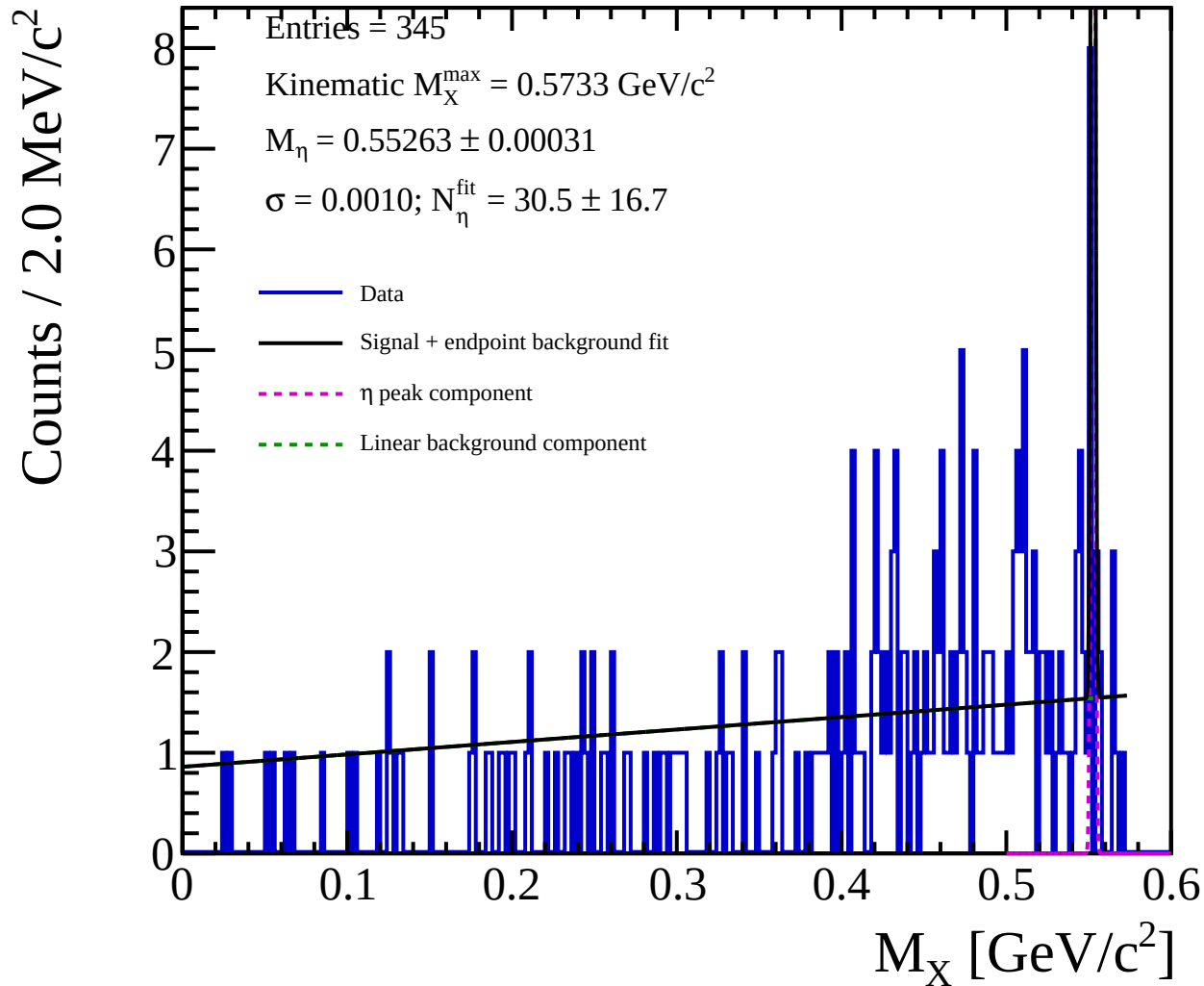
# Production-vertex $K^-$ momentum 768--772 MeV/c



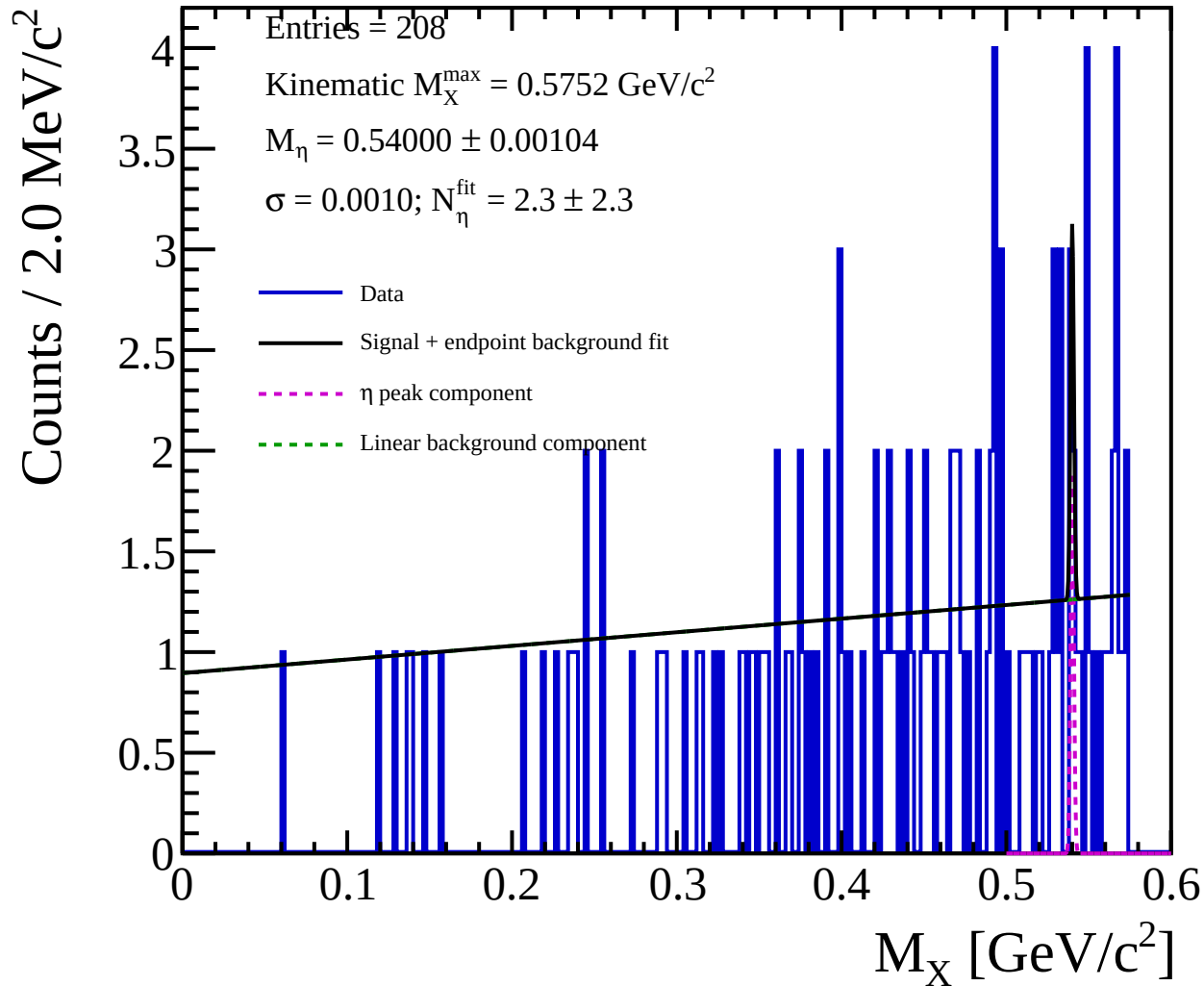
# Production-vertex $K^-$ momentum 772--776 MeV/c



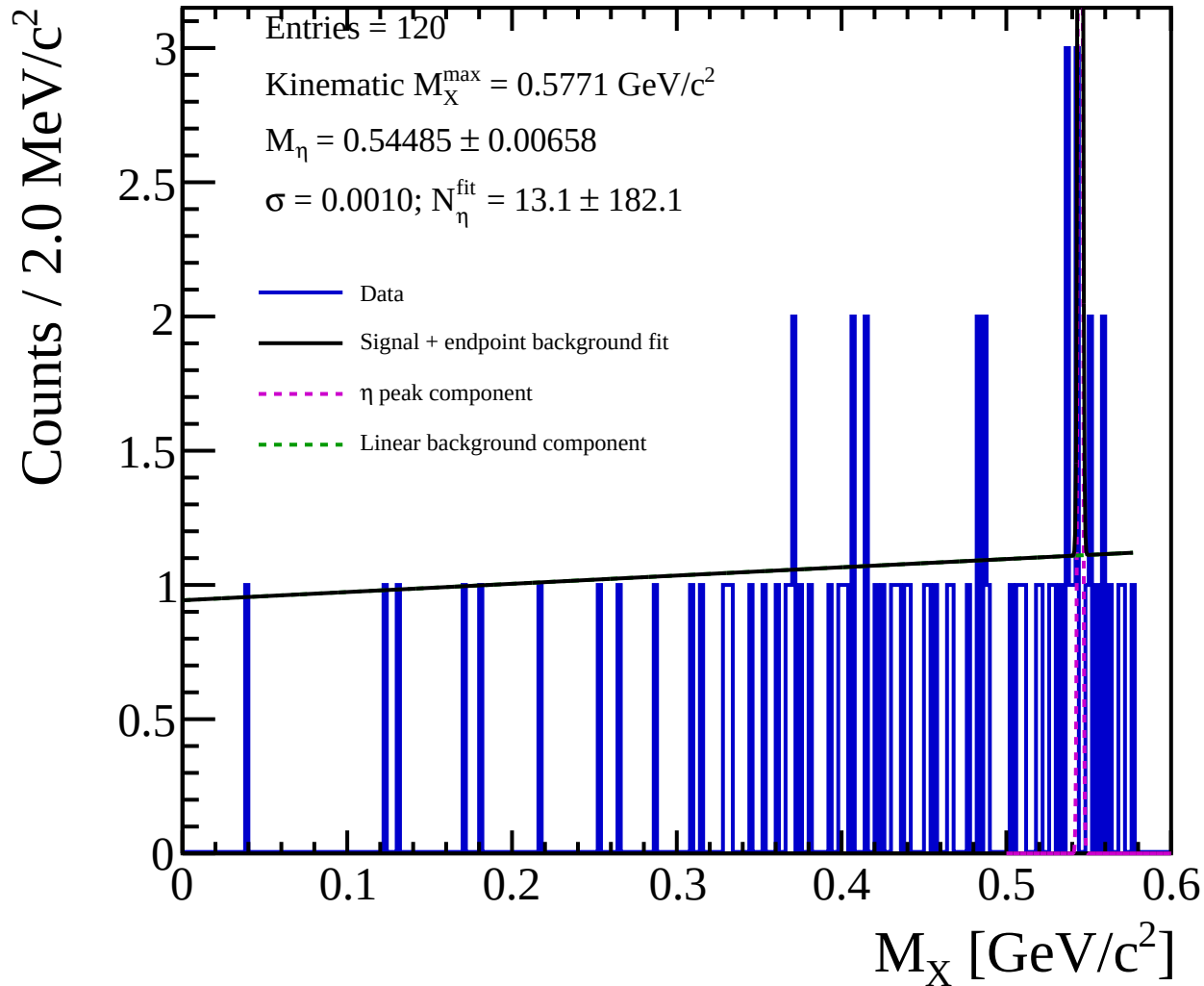
# Production-vertex $K^-$ momentum 776--780 MeV/c



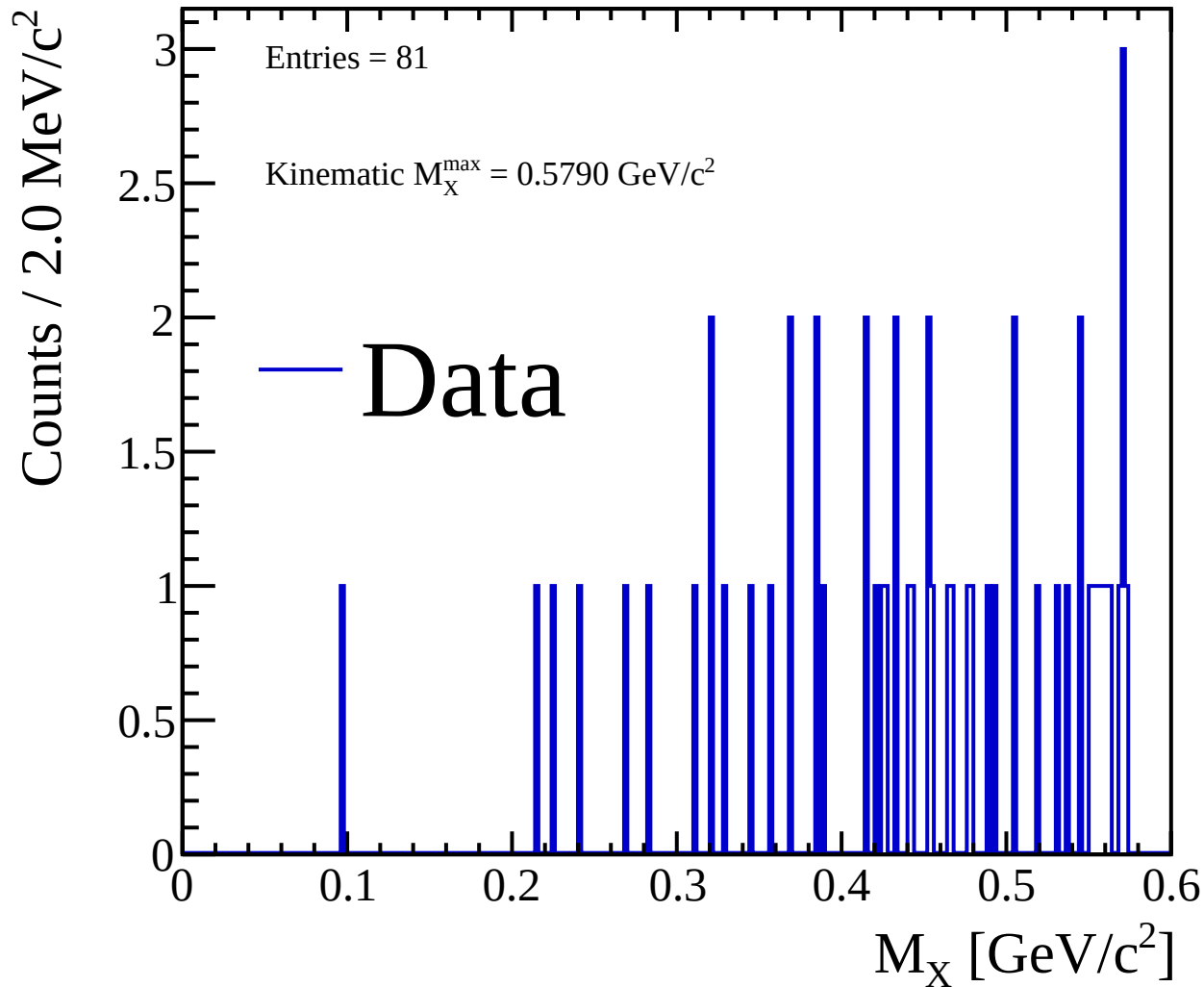
# Production-vertex $K^-$ momentum 780--784 MeV/c



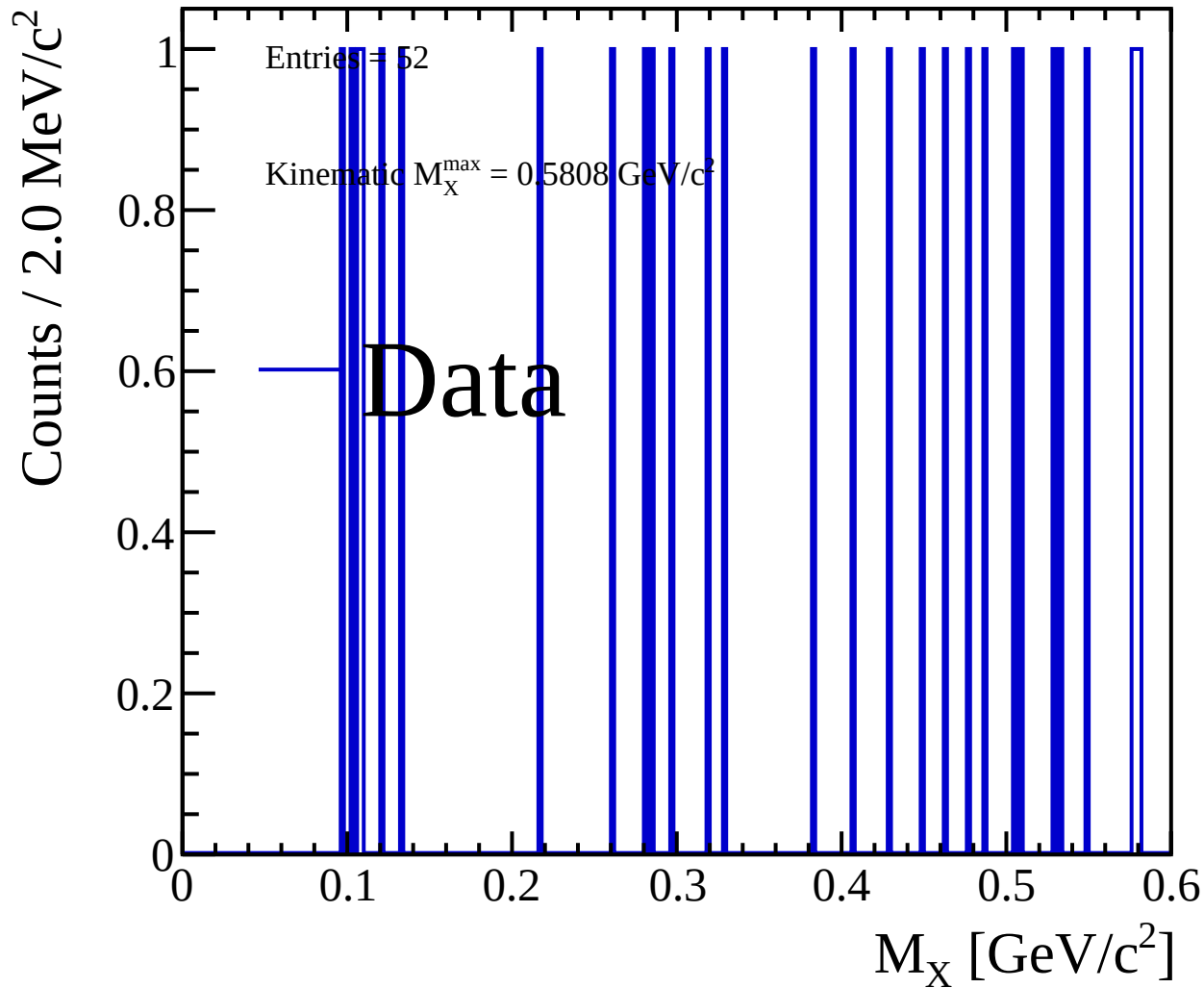
# Production-vertex $K^-$ momentum 784--788 MeV/c



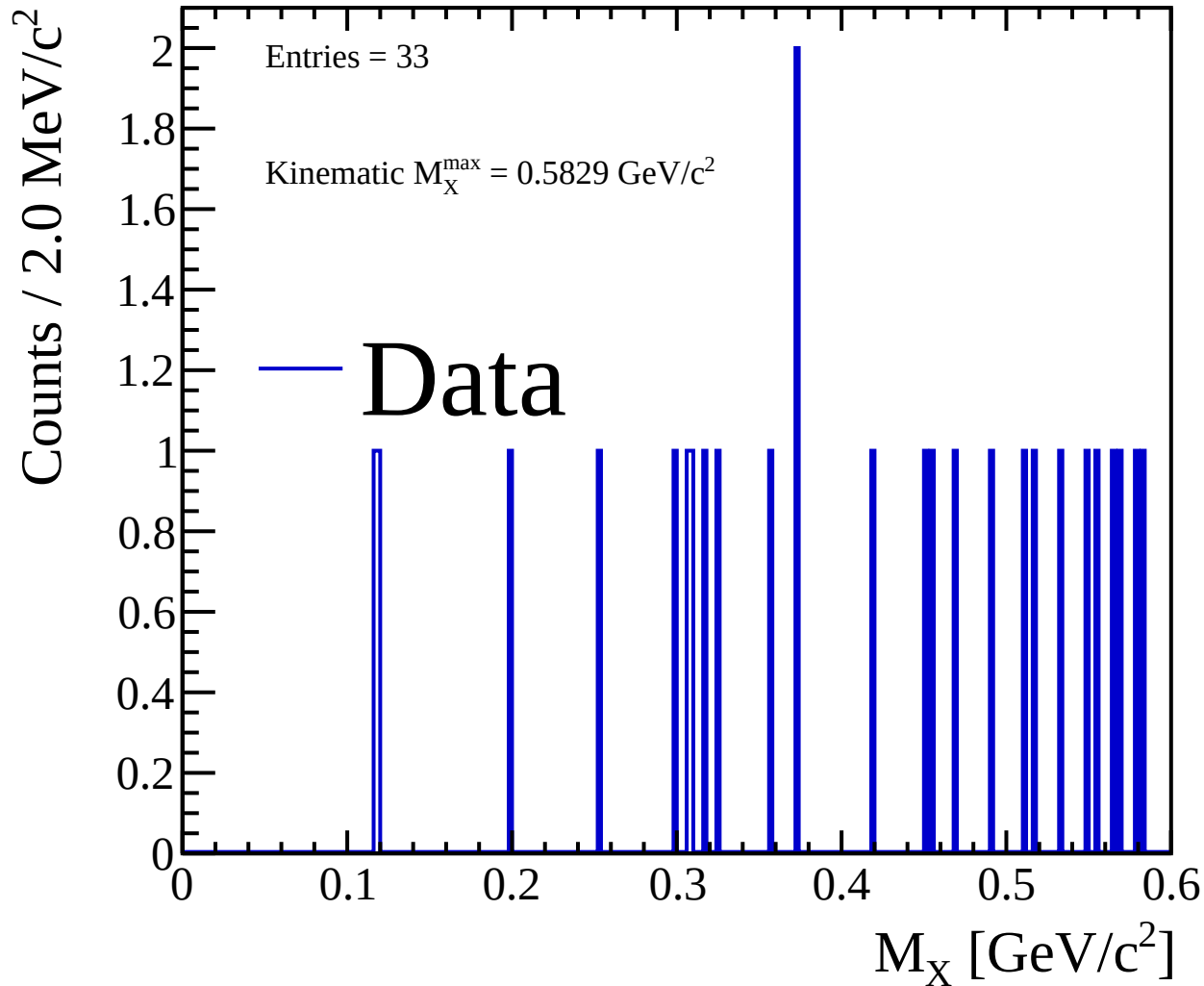
Production-vertex  $K^-$  momentum 788--792 MeV/c



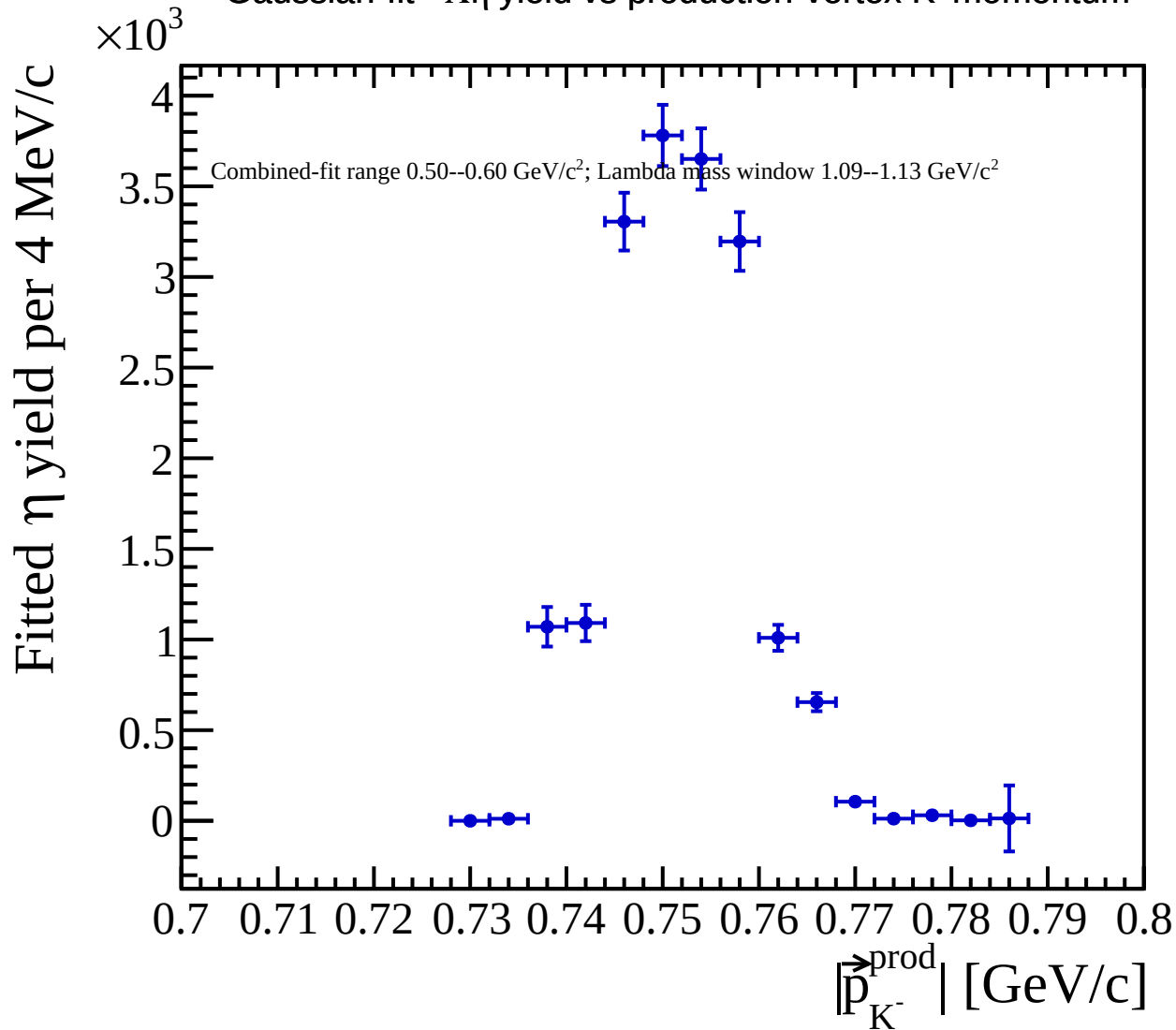
Production-vertex  $K^-$  momentum 792--796 MeV/c



Production-vertex  $K^-$  momentum 796--800 MeV/c



Gaussian-fit  $\Lambda\eta$  yield vs production-vertex  $K^-$  momentum



Fitted  $\eta$  peak position vs production-vertex  $K^-$  momentum

